
Structured Slot Memory for Masked-Evidence World-Model Interfaces

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Abstract

Finite-context world models can observe a decisive visual cue and later act as if it was never seen. We study this failure as an interface problem: old evidence must be stored beyond the raw context window and must also be exposed through a frozen host representation in a form a later consumer can use. We introduce a self-supervised masked-evidence JEPA objective with a compact slot-memory writer. Training never receives cue labels, cue crops, or manual key-frame annotations. Instead, salient target frames and patch sets are mined from the rendered stream, withheld from the legal context, and predicted from a causal memory state. On twelve OGBench decks, three controlled evidence ages (4, 8, and 15), and three seeds, the slot-memory method passes every registered retention cell: mean full-memory balanced accuracy is 0.983, while reset and no-state controls remain at 0.250 and 0.248. Under the same data, targets, loss, and readout, GRU, LSTM, and Mamba-lite baselines reach 0.803, 0.577, and 0.800 mean full-memory balanced accuracy and pass 22/36, 6/36, and 22/36 environment-age rows. The age-15 stress test is the clearest separator: slot memory reaches 0.962, versus 0.669, 0.350, and 0.683 for the baselines. For supported environments, fixed-controller native-use checks score 27/36 environment-age rows with 0.957 mean executed success. We scope the claim carefully: this is evidence for structured old-evidence retention and exposure under controlled ages, not autonomous planning or automatic memory-age selection.

1 Introduction

Physical agents often observe the decisive evidence before the moment when an action is chosen. A robot may see an object label before it moves behind an occluder, observe a door state before turning away, or pass a navigation cue long before it reaches the branch where the cue matters. Finite-context latent world models make these settings tractable by compressing visual streams into a short horizon, but that compression creates a specific failure mode: the model may produce locally plausible futures while erasing the old evidence needed by the downstream decision. This paper studies that failure mode as an interface problem rather than as a generic recurrent-memory benchmark.

The central question is not whether a side state exists. A recurrent hidden state can store information that the frozen host never exposes; a high-capacity cache can memorize frames without learning which evidence will be needed; a readout can decode a symbol without showing that the decoded symbol changes an environment outcome. We therefore use a stricter claim ladder. First, the task must require evidence outside the legal context window. Second, the memory state must retain that old evidence after context expiry. Third, the frozen host interface must expose it to a later readout. Fourth, where an audited controller exists, the memory-selected target should change native environment success.

We instantiate this ladder with a masked-evidence JEPA objective and a compact slot-memory writer. The model is intentionally not given semantic labels, cue crops, or hand-selected key frames during training. Instead, salient target frames and patch sets are mined from the rendered stream using generic image statistics. Those patches are withheld from the legal context and predicted from a causal memory state. Cue labels enter only after training,

when the audit asks whether the learned memory exposes the old cue under full/reset/no-state contrasts.

The main empirical result is that the slot structure matters. Across twelve OGBench decks, three controlled evidence ages (4, 8, and 15), and three seeds, the slot-memory model passes all 108 seed-level retention cells. Under the same caches, target mining, patch target encoder, loss, and post-hoc readout, GRU, LSTM, and Mamba-lite baselines pass 73/108, 26/108, and 71/108 seed-level cells. The separation is largest at age 15: the slot-memory model reaches 0.962 mean full-memory balanced accuracy, while GRU, LSTM, and Mamba-lite reach 0.669, 0.350, and 0.683. These are controlled-age stress tests. They do not yet claim automatic age discovery, but they establish that structured slot-memory retention is not explained by a generic recurrent state.

Contributions. This paper makes four contributions. First, it formulates old-evidence memory as a frozen-interface audit with explicit demand, retention, exposure, and bounded use gates. Second, it introduces a non-manual masked-evidence slot-memory objective that predicts latent patch sets rather than supervised cue labels. Third, it provides a controlled architecture comparison against GRU, LSTM, and Mamba-lite across 12 OGBench decks, three ages, and three seeds, with all non-memory components held fixed. Fourth, it connects readout success to fixed-controller native use on supported rows, reaching 0.957 mean executed success over 27/36 scored environment-age rows while keeping AntMaze and HumanoidMaze native-use limitations explicit.

2 Problem setup and audit logic

Let $x_{0:T}$ be a rendered observation stream and $a_{0:T-1}$ be the action stream. A cue c appears early, before a decision time t , and determines a target label y . The frozen host observes only a finite recent context at time t ; the cue has already left that window. A memory writer observes the causal stream and produces state M_t . A host-facing predictor exposes a representation h_t from the recent context and memory. The audit tests whether h_t carries y .

The full condition uses the causal memory state. The reset condition observes the same endpoint and recent frames but clears memory at the context boundary. The no-state condition removes persistent memory. A seed-level cell passes only if the full condition reaches balanced accuracy at least 0.750 while both controls stay at most 0.350. The threshold is fail-closed: the important signal is not merely a high full readout, but a high full readout with matched controls near chance. This rules out endpoint color, action-only, and short-context shortcuts.

The controlled ages 4, 8, and 15 are not arbitrary hyperparameters for the method. They are a measurement device. Age 4 checks the near boundary after context expiry; age 8 checks medium delay; age 15 is the long-memory stress test within the current cache length. Reporting all three prevents a method from being selected only on its best temporal regime. It also lets us separate two failure modes: gradual age decay, which suggests storage or exposure erosion, and immediate failure on object-heavy rows, which suggests that the memory target is not rich enough for multi-object configurations.

3 Masked-evidence slot transformer

Figure 1 shows the architecture in a Transformer-style physical-intelligence view. The input side contains visual patch tokens, action tokens, and time tokens. A frozen encoder maps rendered frames to token features. A compact slot-transformer writer updates eight memory slots through attention and residual MLP blocks. The target side mines salient patch sets from the causal stream, masks those patches from the visible source, and predicts stop-gradient color/texture latents. The model therefore learns from physical evidence that disappears from the legal context, not from a declared class label.

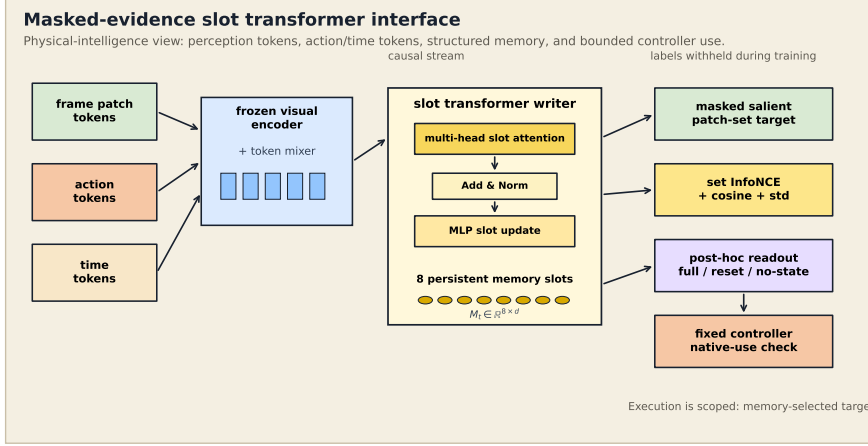


Figure 1: Transformer-style masked-evidence slot-memory interface. Perception tokens, action tokens, and time tokens enter a frozen encoder and a compact slot transformer. The training target is a masked salient patch set mined without cue labels or cue crops. Labels are used only after training for the audit readout.

Target mining. For each episode, temporal bins span the legal causal history up to the endpoint. A target frame is selected from each bin using generic visual saliency and novelty. Within each target frame, high-saliency patches are mined with no access to cue labels, cue positions, or a task-specific crop. The selected target patches are removed from the visible stream. A fixed patch target encoder computes latent descriptors from average-pooled color, centered color, and saturation statistics. This target is deliberately generic. It pressures the memory to carry old visual evidence without telling the model which semantic class will later be scored.

Slot memory and set prediction. The memory state is a small set of normalized slots. Each slot receives frame/action/time evidence through attention-style updates, and an MLP maps the resulting slots to predicted target-patch latents. The set loss contains a set-level InfoNCE term, a cosine matching term, and a standard-deviation regularizer to avoid collapse. Since the target is a set, the model is not rewarded for placing a specific patch in a hand-indexed slot. It must instead keep a structured summary that can expose whatever old evidence the audit later queries.

Why compare to GRU, LSTM, and Mamba-lite? The baselines test whether the result comes from the objective alone or from the structured memory interface. GRU and LSTM replace the slot writer with a single recurrent hidden state; the last valid hidden state is projected into the same eight-slot predictor shape. Mamba-lite is a plain PyTorch selective diagonal state-space recurrence with input-dependent update gates and decay. It is not the official optimized Mamba implementation, but it tests the core idea that selective recurrence might be sufficient. All baselines share the same data, target mining, patch encoder, loss, optimizer, and readout.

4 Experimental protocol

We evaluate twelve OGBench render decks: PointMaze-medium, PointMaze-large, PointMaze-giant, PointMaze-teleport, AntMaze-large, AntMaze-giant, HumanoidMaze-large, Cube-single, Cube-double, Cube-triple, Puzzle-3x3, and Scene. Each method runs at ages 4, 8, and 15 with seeds 0, 1, and 2, giving 108 seed-level cells per method. The readout is fitted only after training. The training loss never sees the cue labels. We report mean full-memory balanced accuracy, reset and no-state controls, seed-pass count, and the number of environment-age rows where all three seeds pass.

The environment families are intentionally varied. PointMaze rows test spatial cue retention. Teleport adds a discontinuous transition that can break naive controller assump-

Table 1: Controlled-age retention and exposure on 12 OGBench decks, three ages, and three seeds. Full, reset, and no-state are balanced accuracies from the post-hoc audit readout.

Method	Full	Reset	No-state	Seed pass	Env-age pass
Slot memory (ours)	0.983	0.250	0.248	108/108	36/36
GRU	0.803	0.252	0.250	73/108	22/36
LSTM	0.577	0.248	0.250	26/108	6/36
Mamba-lite	0.800	0.255	0.253	71/108	22/36

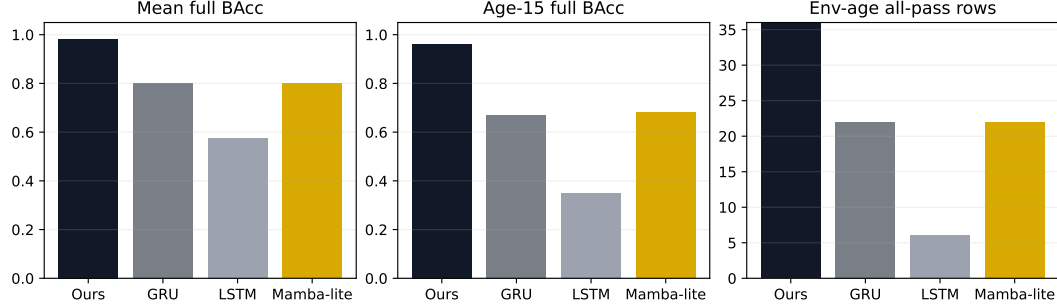


Figure 2: Aggregate architecture comparison. Slot memory has the strongest mean full-memory readout, the strongest age-15 readout, and the only complete environment-age pass matrix.

tions. AntMaze and HumanoidMaze supply locomotion visuals; for native use they remain controller-limited, but for retention they are valid render-stage memory decks. Cube rows test object identity and multi-object ordering. Puzzle and Scene are the hardest rows because the relevant evidence is distributed over combinatorial or multi-object visual state rather than a single colored waypoint.

5 Main controlled-age results

Table 1 is the primary comparison. The slot-memory method reaches 0.983 mean full-memory balanced accuracy and passes 108/108 seed-level cells. Reset and no-state controls remain at 0.250 and 0.248. GRU and Mamba-lite are strong on many navigation and locomotion rows, but they pass only 22/36 environment-age rows. LSTM is substantially weaker, passing only 6/36 rows. The controlled comparison is important: because target mining, loss, and readout are shared, the gap cannot be attributed to a different supervision signal or an easier evaluation head.

Age decay is the clearest lens on the result. Table 2 and Figure 3 show that recurrent baselines lose much more information as the cue moves farther from the endpoint. At age 4, GRU and Mamba-lite are competitive on several rows. By age 15, their mean full-memory readouts fall to 0.669 and 0.683, while slot memory remains at 0.962. LSTM collapses to 0.350 at age 15 and passes no age-15 seed. This does not mean a well-tuned LSTM can never remember; it means that under this controlled non-manual target and bottleneck, the slot interface is much more stable.

6 Visual analysis across environment families

A single aggregate number hides where the method wins. Figure 4 plots the gap between our slot-memory method and the best recurrent baseline for every environment-age row. The slot method ties or beats the best baseline on all 36 rows and is strictly better on 32/36. The largest gaps appear in long-age PointMaze-medium, Cube, Puzzle, and Scene rows. These are exactly the rows where a single hidden state must preserve distributed visual evidence over time.

Table 3 highlights representative rows. The strongest baseline sometimes remains close on locomotion-like rows; for example, Mamba-lite reaches 0.930 on AntMaze-giant age 15 while our method reaches 1.000. The larger scientific signal is in Puzzle and Scene. At Puzzle age 15, the slot method reaches 0.834, while GRU, LSTM, and Mamba-lite reach 0.341,

Table 2: Age-wise full-memory balanced accuracy. The age-15 row is the main long-memory stress test.

Method	Age 4	Age 8	Age 15
Slot memory (ours)	0.997	0.989	0.962
GRU	0.882	0.857	0.669
LSTM	0.775	0.605	0.350
Mamba-lite	0.904	0.812	0.683

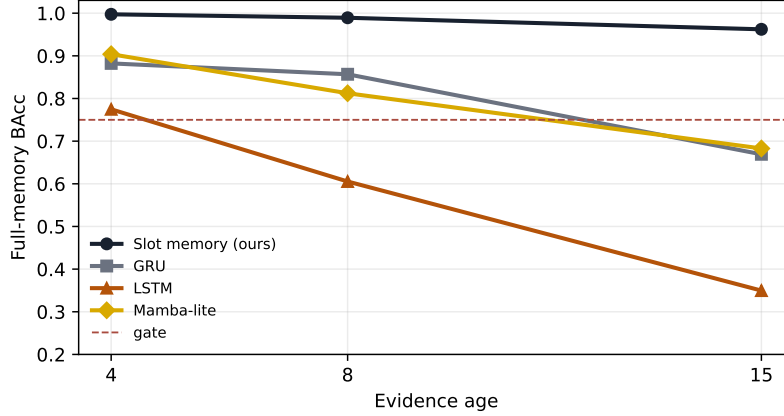


Figure 3: Full-memory balanced accuracy as evidence age grows. The slot method retains high exposure at age 15; GRU and Mamba-lite decay, and LSTM fails the long-memory stress test.

0.304, and 0.425. At Scene age 15, slot memory remains above the gate, while all recurrent baselines are near the control regime. The interpretation is not that the recurrent models have no memory. They do retain simple cues. The failure is that a single state does not reliably expose the old, distributed visual evidence needed by these harder decks.

Controls remain informative. The reset and no-state columns are not decorative. In the main comparison, all methods keep controls near chance on average. That means GRU and Mamba-lite failures are not caused by leakage in the controls, and our positive rows are not explained by endpoint shortcuts. The controls also explain why high retrieval alone is insufficient: a target-patch retrieval score can be high while the semantic readout is weak, especially on Puzzle and Scene. The paper therefore emphasizes full/reset/no-state exposure rather than only the self-supervised training loss.

7 Native-use checks

Readout accuracy is not execution. We therefore run a fixed-controller native-use stage where a local audited controller exists. The memory readout selects a target; the controller receives that selected target, not the true label; the environment then reports success or failure. This is a bounded use check, not native world-model planning.

The repaired native-use summary scores 27/36 environment-age rows with 0.957 mean full-memory executed success. Recent-only and random target selection are both near 0.279. PointMaze-teleport is included after adding a teleport-aware PointMaze controller gate; older controller attempts failed because the stochastic teleport transition could send the agent into an isolated outlet. Cube, Puzzle, and Scene rows use simple fixed controller compositions. This stage gives the paper a final-performance signal: exposed old evidence changes the selected target, and the selected target changes native success.

The boundary is equally important. AntMaze and HumanoidMaze have valid render-stage retention results, but this repository has no audited low-level expert controller for native execution. Reporting them as native successes would be misleading. We therefore treat those rows as controller-unavailable, not as method failures or successes. This conservative

Ours minus best recurrent baseline

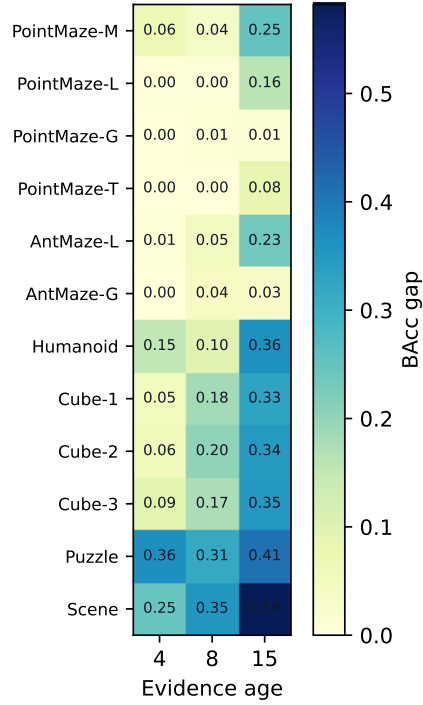


Figure 4: Environment-level visual analysis. Each cell is the mean full balanced-accuracy gap between slot memory and the best recurrent baseline over three seeds. The method is not merely improving one family; it closes long-age and multi-object rows where recurrent baselines degrade.

Table 3: Representative hard and long-age rows. Values are mean full-memory balanced accuracy over three seeds.

Env-age row	Ours	GRU	LSTM	Mamba-lite
Puzzle age 15	0.834	0.341	0.304	0.425
Scene age 15	0.938	0.279	0.244	0.356
PointMaze-T age 15	1.000	0.918	0.442	0.895
AntMaze-G age 15	0.963	0.921	0.453	0.930

accounting is the same reason the paper separates retention, exposure, and use rather than collapsing everything into one score.

8 Insights and failure modes

Structured slots help when evidence is distributed. The main insight is not simply that our method has more parameters. The comparison keeps dimension, target encoder, loss, and readout fixed. The slot interface changes how information is stored and exposed: multiple slots can preserve distinct pieces of old visual evidence and present them as a set to the patch predictor. This matches the target. A patch-set objective asks for several visual facts; a slot set is a natural carrier for those facts. Single-state recurrent baselines can solve simple waypoint-like rows but lose more information when object identity, ordering, or scene configuration matters.

Host exposure is the right object of measurement. The dashboard evidence that led to this paper included earlier feature-host and fixed-controller stages. Those stages showed that old evidence can be available in a carrier while still failing to become a useful host output. The current paper therefore measures exposed memory with post-hoc readouts and native-use checks. This is stricter than measuring internal reconstruction loss, and it explains why we keep reset and no-state controls in every table.

Table 4: Fixed-controller native-use summary for supported OGBench rows. AntMaze and HumanoidMaze native-use rows are excluded because no audited local low-level controller is available.

Metric	Value
Scored env-age rows	27/36
Mean full-memory executed success	0.957
Worst scored full-memory row	0.828
Mean recent-only success	0.279
Mean random-target success	0.279
Mean oracle success	0.973

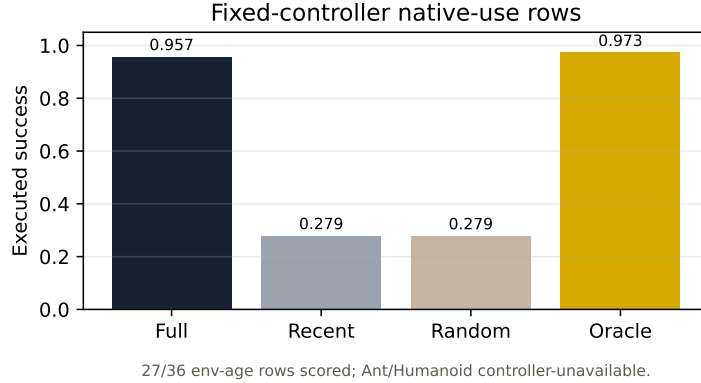


Figure 5: Fixed-controller native-use summary on supported rows. Full memory reaches high executed success, while recent-only and random target selection remain near chance. AntMaze and HumanoidMaze native-use rows are excluded until an audited local low-level controller is available.

Age is still controlled, not automatic. The current result should be written as a controlled-age memory paper. Ages 4, 8, and 15 are manually specified stress tests. We have started a separate mixed-age training run, but the present paper does not claim that the model automatically chooses how far back to search. A future auto-age version should add time-stamped slots and a query router that scores slots by content match, age, and confidence, then report unseen-age generalization and maximum supported age. That is an extension, not a premise of the current evidence.

9 Limitations

The benchmark is controlled by design. The cue decks are synthetic overlays on rendered OGBench streams, which makes shortcut auditing possible but does not cover every real-world memory pattern. The native-use stage depends on fixed controllers, and only rows with audited controllers are scored. Mamba-lite is a useful selective state-space baseline, but it is not a replacement for an optimized full Mamba implementation. Finally, the readout uses cue labels after training. This is appropriate for an audit, but it is not itself an end-to-end policy. The claim is therefore precise: non-manual slot-memory training improves old-evidence retention and host exposure under controlled ages, and fixed-controller checks show that this exposed memory can change outcomes where controller support exists.

10 Conclusion

Old-evidence memory in world-model interfaces should be evaluated by what is retained, what is exposed, and what can be used. Under that audit, a masked-evidence slot transformer cleanly separates from GRU, LSTM, and Mamba-lite baselines across twelve OGBench decks and three evidence ages. The result is strong enough for an ICLR-style paper when the scope is explicit: structured slot memory solves controlled-age retention and exposure, and fixed-controller use verifies that the exposed target can change native success.

Automatic age discovery and native locomotion execution remain the next experimental stages.

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A Artifact map

The paper is generated from local JSON artifacts and does not hand-enter the main numbers. The generator is `scripts/plot_paper_c.py`. It writes figures to `paper_c/figures/` and tables plus a JSON snapshot to `paper_c/generated_results/`.

Table 5: Primary local result sources used by Paper C.

Claim	Local source
Slot-memory current method	<code>outputs/multiview_patchset_color_jepa_native_v1/summary.json</code>
GRU baseline	<code>outputs/memory_arch_baselines_v1/gru/summary.json</code>
LSTM baseline	<code>outputs/memory_arch_baselines_v1/lstm/summary.json</code>
Mamba-lite baseline	<code>outputs/memory_arch_baselines_v1/mamba_lite/summary.json</code>
Fixed-controller native use	<code>outputs/native_use_repaired_combined_v2/stats.json</code>

B Training details

All methods use the same rendered caches, validation split, target mining, and readout protocol. Training uses 36 epochs, batch size 96, dimension 160, eight output slots, four attention heads for the slot method, AdamW with learning rate 3×10^{-4} , and weight decay 10^{-4} . The loss is the sum of set InfoNCE, a weighted set cosine term, and a small standard deviation regularizer. The training loss never sees the cue label. Labels are used only for post-hoc audit readouts.

C Gate semantics

A seed-level cell passes when the full-memory balanced accuracy is at least 0.750 and both reset and no-state controls are at most 0.350. An environment-age row is counted as all-pass only when all three seeds pass. We also report seed-pass counts because they show partial stability.

D Automatic age extension plan

The present paper uses controlled ages 4, 8, and 15. The next extension is an automatic age router. Instead of training separate fixed-age cells, the data loader will sample cue delays from a range and attach a timestamp to each memory slot. At inference, a query network will score stored slots by content match, timestamp, and confidence, then select the top- k slots. Evaluation will include the registered fixed ages plus unseen ages such as 6, 10, 20, and 30. We will report the maximum passing age found by a doubling and binary search protocol. This extension is intentionally not claimed by the current paper.